

A claim about a population

A hypothesis test uses sample evidence to assess a claim about a population parameter.

Does a claim about average processing time concern μ or the observed sample mean?

The null hypothesis

The null hypothesis specifies the population claim assessed as the reference model.

For a claimed mean processing time of 12 minutes, how could we write H_0 ?

The alternative hypothesis

The alternative describes the population change or difference for which we seek evidence.

If the question is whether the mean exceeds 12 minutes, how should H_a be written?

Direction of the question

A one-sided alternative specifies a direction; a two-sided alternative allows either direction.

Which alternative matches a question about any change from $\mu=12$?

Choosing before looking

The research question should determine the alternative before inspecting the outcome.

What problem arises if we choose the tail only after seeing whether the sample mean is high or low?

The null sampling model

Under H_0 , a sampling model describes the statistic's variation if the null claim holds.

Why should this model concern sample means rather than individual processing times?

A test statistic

A test statistic summarizes how far sample evidence departs from the null model.

Why should distance from the null value be judged relative to standard error?

An assumption check

A valid test relies on the sampling design and the assumptions of its probability model.

Could a very small p-value repair a strongly biased sample?

The p-value

A p-value is the null-model probability of results at least as extreme as observed, in the direction of the alternative.

Which assumption about H_0 is used when calculating it?

One tail or two?

Synthetic result: the sample mean is above the null mean. The alternative is two-sided.

Would considering only equally high sample means capture all results at least as extreme?

A p-value is not a hypothesis probability

Synthetic result: a valid test reports $p\text{-value}=0.03$.

Does this assign a 3% probability to H_0 being true?

Significance level

The significance level α sets the rejection rule and controls the Type I error rate under a valid test.

Why should α be chosen before examining the p-value?

A rejection decision

Synthetic valid test: $p\text{-value}=0.03$;
significance level $\alpha=0.05$.

What decision follows for H_0 ?

A nonrejection decision

Synthetic valid test: $p\text{-value}=0.18$;
significance level $\alpha=0.05$.

Does failing to reject H_0 establish that it is true?

Type I error

A Type I error rejects a null hypothesis that is true.

If H_0 states that mean processing time is 12 minutes, what would a Type I error conclude?

Type II error

A Type II error fails to reject a null hypothesis that is false.

If the true mean exceeds 12 minutes, what would a Type II error miss?

Power

Power is the probability of rejecting H_0 for a specified alternative that is true.

How might a larger sample help detect a small departure from the null value?

Statistical and practical importance

A small p-value measures evidence against a null model, rather than the size or importance of the difference.

Could a 0.02-minute difference be statistically significant but operationally unimportant?

An interval connection

For matching two-sided mean procedures, a 95% confidence interval excluding μ_0 corresponds to rejecting $H_0: \mu = \mu_0$ at $\alpha = 0.05$.

Would a matching interval from 12.2 to 12.8 reject a null mean of 12?

A careful conclusion

Synthetic valid test rejects $H_0:\mu=12$ in favor of $H_a:\mu>12$.

How would you state the evidence without claiming proof or explaining why the mean changed?